

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY (TET)

FINAL YEAR PROJECT GUIDELINE

(NTA-06 & NTA-08)

NOVEMBER 2022

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CHAPTER ONE: INTRODUCTION

The National Institute of Transport (NIT) is a public higher learning institution that was established in 1975 and in 1982 NIT was re-established through NIT Act No 24 of 1982 as an autonomous Higher Learning Institution under the then Ministry of Communications and Works.

NIT is the only such training Institute in Sub-Saharan excluding South Africa, mandated by the Government of United Republic of Tanzania to offer training in transport and communications.

The Institute offers programs in Automobile Engineering, Mechanical Engineering, Air craft Maintenance Engineering, Electronics and Telecommunication Engineering , Electrical Engineering, Auto-Electrical and Electronics Engineering, Telecommunication and Railway Signaling Engineering, Ship Building and Repair , Naval Architecture and Marine Engineering, Pipe works, Oil and Gas Engineering, Electrical and Railway Electrification Engineering, Civil and Transportation Engineering, Logistics and Transport Management, Freight Clearing and Forwarding, Procurement and Logistics Management, Business Administration, Computing Information Communication Technology and Human Resources Management. NIT also is offering Post Graduate Diplomas in Air Transport Management, Logistics and Transport Management, Travel and Tourism, Shipping and Port Management, Procurement and Logistics Management, Rail Transport Management, Transport Economics, Records, Archives and Information Management Road Transport Safety Management and Transport Engineering Management. NIT also offers short term programs and provides Research, Consultancy and Community Services.

The Institute is fully accredited with the National Council for Technical Education (NACTE) to provide Competence Based Education Training (CBET) programs at the level of Certificate, Diploma and Degree (National Technical Awards (NTA) level 4 to 8). NIT has mandate to conduct its own examinations and grant own awards as approved by National Council for Technical Education (NACTE)

Assessment is done through written Examinations, Research, practical and project works to assess knowledge, skills, understanding and wider attributes acquired by

students at NIT within their learning period. Student enrolled under the Transport Engineering and Technology Department, Project work or research is one of the modules for the fulfilment of the award of Diploma or Bachelor in Automobile Engineering, Mechanical Engineering, Air craft Maintenance Engineering, Electronics and Telecommunication Engineering , Electrical Engineering, Auto-Electrical and Electronics Engineering, Telecommunication and Railway Signaling Engineering, Ship Building and Repair , Naval Architecture and Marine Engineering, Pipe works, Oil and Gas Engineering, Electrical and Railway Electrification Engineering, Civil and Transportation Engineering.

The Transport Engineering and Technology (TET) department Project guideline, guides students and Supervisors on how to carry out and supervise project works respectively. There are two categories of project namely Project I and Project II for Bachelor degree and ordinary diploma programs. Project Work I is carried out during the seventh semester and fifth semester for both levels respectively while Project Work II is carried out during the eighth and the sixth semester of the programme semester for both levels respectively.

Project I and II each contributes 12 credits. The Project guideline for TET has been prepared according to the curriculum which has been already approved by NACTE.

CHAPTER TWO

2.0 Project Work Administration

The Head of Department in collaboration with the Overall Project Coordinators and supervisors (appointed by the Head of Department) will administer the project work.

2.1 What is a Project Work?

Project work is a task on a subject selected by a student or recommended to the student by a supervisor involving application of skills and knowledge towards solving engineering and or community problem(s) analytically, experimentally and design or a combination of the mentioned methods. Application of knowledge acquired at the Institute is a prerequisite.

2.2 Objectives

The need in engineering education for the combination of theory and practice has led to the decision to give emphasis on problem-based teaching and learning at NIT particularly in the Transport Engineering and Technology Department. The final year project is thus designed to be problem-based.

Objectives of Engineering Projects are:

- a. To provide an environment in which students gets an opportunity to train in and assimilate to the vision and mission of the Institute in a broader outlook.
- b. Develop the student's capability for self-reliant solving of engineering problems
- c. Extend the students' knowledge considerably in their chosen topic
- d. Develop the skills required for originality and creativity of thinking
- e. Contribute to the existing body of knowledge of engineering
- f. To contribute to the fulfillment for the award of Ordinary Diploma and Bachelor degree in Engineering.

2.3 Outcomes

As a part of outcomes, student should be able to:

- a. Identify and define an engineering problem clearly and specify conditions and limitations of solving problem.
- b. Plan and collect independently relevant information for the solution of the problem.
- c. Apply knowledge acquired in the course of study to develop a suitable solution to the problem.
- d. Select an optimal solution to the problem while taking into account all practical engineering aspects, such as economical, ecological and other related point of view.
- e. Present orally and in a well written report ideas, suggestions, criticisms, conclusions and recommendations of the findings of the projects.
- f. Successfully defend the project in front of team of Examiners.

2.4 Skills gained

In addition to academic and personal rewards, the project will allow students gain the following skills:

a. Self management

- i. Set personal objectives
- ii. Manage time and tasks
- iii. Identify and implement personal learning strategies

b. Problem solving

- i. Analyse theories and evidence
- ii. Think conceptually and identify strategic options
- iii. Evaluate the usefulness of different approaches
- iv. Develop skills towards finding solution independently

c. Information Technology (IT)

- i. Use research/ project tools
- ii. Practice information retrieval

d. Learning skills

- i. Learning independently
- ii. Use library skills and other information gathering skills
- iii. Intergrade knowledge from various sources

e. Communication

- i. Write fluently and effectively on chosen topics
- ii. To interpret engineering drawing
- iii. Understand and Present coherent argument orally and in writing

f. Project management

- i. Prepare activity plan.
- ii. Write a professional engineering report

2.5 Project Type, Title and Specifications.

- i. Project taken by the students fall into, but not limited to the following categories:
 - a. Product type
 - b. Research type
 - c. Problem solving type.
 - d. Experimental type
 - e. Case study.
- ii. To select a project type, students should first study the community and the environment surrounding them physically, identify core problems in the society or industry then suggest practically appropriate title. The title chosen should reflect the problems identified. Any project undertaken by a student should:
 - a. Always have a link to engineering or technological problems in communities
 - b. Require application of knowledge and skills acquired in the course of study.
 - c. Facilitate a student to handle industrial machinery and processes.
 - d. Require students to implement own ideas, designs, findings etc. in order to solve the identified problem.

- e. If possible, be connected to students' projects of other Engineering departments.
 - f. Be limited in time according to the time allocation as per project schedule.
- iii. Each student should be prepared to present his/her proposal in front of departmental panel of academic staff that deliberate, amend and approve or disapprove it. Student's presentations should follow the timetable given.

2.6 Project Levels

Project should be divided into two main parts as project I and II which will be conducted in semester I and II respectively.

2.6.1 Project Work I

This part should involve the introductory lectures and writing the proposal.

a. Introductory Lectures

Introductory lectures on project formulation and details on all project guidelines conducted by departmental project coordinator. The duration for the above lectures should be one (1) or two (2) in the beginning of the semester.

b. Project Concept Note

Project concept note presentation (Max 3.pgs)

Contents:

1. Title of the project
2. Problem Statement
3. Project objective
 - 3.1.Main objective
 - 3.2.Specific objective
4. Significance of the project
5. Project cost estimations
6. Project Schedule

c. Project Proposal

The project proposal will state directly the following:

- i. Title of project
- ii. Definition of the problem
- iii. Project objectives
- iv. Significance of the project
- v. Literature review
- vi. Methodologies (Materials and Methods) of solving the problem
- vii. Cost estimate
- viii. Gant chart (schedule of activities)
- ix. References.
- x. Appendixes

2.7 Preparation of project proposal

2.7.1 Title of project

Project Title is a name of the Project. A proper project title describes the whole assignment in one sentence. Project titles enables students to understand the main goal of the Project work and deliverables. Titles must be limited to 15 words, be attractive, exciting and must convey the meaning, the area of intervention and the goals of the project which include prototype implementation.

2.7.2 Choice of Project

The student should start thinking about possible topics as early as possible and undertake a relevant literature review to see how problems are framed and what arguments are made. The student should bear in mind that the topic must be intellectually feasible, practicable in terms of gaining access to the necessary sources and manageable in the time available. Also, the topic must be related to the area of study taken by a student.

2.7.3 Definition of the problem

Problem statement is the description of an issue currently existing which needs to be addressed. It provides the context for research/project study and generates the

questions which the research/project aims to answer. The statement of the problem is the focal point of any research project. A good problem statement is just one sentence with several paragraphs of elaboration.

2.7.4 Project objectives

A project objective describes the desired results of a project, which often includes a tangible item. An objective is specific and measurable, and must meet time, budget, and quality constraint. The objectives should be achievable in a stated academic year of study.

2.7.5 Significance of the project

The significance of the project/ research work is one section of a project that tends to showcase the importance of the project or research work. When writing Significant of the project, the researcher should be able to enumerate all the benefits of the project. The sole aim of Significance of the project/ research is to show at a glance the contribution of project/ research to the body of existing or nonexistent knowledge or give your Supervisor/Reader an insight on how well the project/research will be of benefit to them.

2.7.6 Literature Review

A literature review is a comprehensive summary of previous project/ research on a topic. The literature review surveys scholarly articles, books, and other sources relevant to a particular area of project / research. The review should enumerate, describe, summarize, objectively evaluate and clarify this previous project/research. It should give a theoretical base for the research and help (the author) to determine the nature of research. The literature review acknowledges the work of previous researchers, and in so doing, assures the reader that your work has been well conceived. It is assumed that by mentioning a previous work in the field of study, that the author has read, evaluated, and assimilated that work into the work at hand.

2.7.7 Methodology

This part Provides a detailed description of how the stated objectives of the project will be achieved. Include a detailed outline of methods used in your analysis and validation. This section of the proposal is very important and should provide a complete picture of how the intended work will be accomplished.

2.7.8 Cost estimates

Cost estimation in project management is the process of forecasting the financial and other resources needed to complete a project within a defined scope.

Note that: The institute does not provide any fund to student for their projects nor the department, even though project is a major part of an academic year that a student practice execution of tasks within their proposed budget. Therefore, Students will be required to carry out purchases of components not found in the laboratories or workshops.

2.7.9 Gantt chart (Schedule of activities)

A Gantt chart, commonly used in project management, is one of the most popular and useful ways of showing activities (tasks or events) displayed against time. On the left of the chart is a list of the activities and along the top is a suitable time scale.

2.7.10 Reference

Extensive information gathered, showing impressive depth and breadth of reading, used effectively to support the work. The information is correctly and accurately referenced using the appropriate conventions. For more information refer to section 3.1.7 “*Referencing Citation and Caption*”

2.8 Project Work II

This part should involve the experiment based on project one to completion of the project report; it includes:

- a. Practical's (Materials and Methods)
- b. Building of prototype
- c. Data analysis (Testing procedures) and discussions (results)
- d. Conclusion and recommendations
- e. Report writing

2.9. Description of chapters in Project proposal / Project proposal outline

2.9.1 Chapter one: Introduction

2.9.2 Chapter two: Literature review

2.9.3 Chapter three: Methodology

2.9.4 References (according to guideline requirement)

CHAPTER THREE:

3.0 Preparation and Submission of Project Report

Candidates must follow the specifications laid out in this guideline when preparing their project report. **Candidates who will NOT follow specifications given in this document; their project report will not be accepted.**

3.1 Preparing the Project Report

3.1.1 Length of project report

- i. Project report should not be less than 40 pages excluding appendices.
- ii. Candidates should use gender neutral language in writing their project report unless the nature of the project requires the use of gender specific language.

Note: avoid any form of personalization i.e. I, you, we etc.

- iii. All project reports shall be written in English Language, specifically British English. Other English forms or standards are not accepted.
- iv. The project report must be printed on A4 (210 x 297 mm), acid free white quality paper and printed on single-sided right-hand pages only

3.1.2 Preliminary Pages Pagination

- a. All pages in the preliminary section such as Acknowledgements, Table of Contents etc. should be numbered in consecutive lower-case Roman numerals (i, ii, iii, iv, etc.), except for the Title page which is suppressed.
- b. All pages of the main text should be numbered consecutively in Arabic numbers,
- c. All references or bibliography, tables and figures should be numbered consecutively (chapter wise) in Arabic numerals (1, 2, 3, etc.). i.e. Figure 2.1, Figure 3.4
- d. Page numbers must not be punctuated, bracketed, hyphenated nor accompanied by any decorative symbols and should be bottom centred. The position of the page number must be consistent throughout the project report.
- e. Each appendix should be identified separately using a capital letter (APPENDIX A, APPENDIX B, etc.) on the top centre of the page. **No separator pages are allowed.**

3.1.3 Typing

Text must be in the following format:

- a. Type written/word-processed
- b. In a font of 12 points (Times New Roman) and Text should not be scripted or italicised except for: Scientific names, terms in a different language, equations and quotation.
- c. The text should be in 1.5 line spacing and have a font size of 12 using Times New Roman font. Double spacing should be provided above and below equations.
- d. Never use bullets in numbering system.

3.1.4 Paragraphs

- a. Paragraphs should not be indented.
- b. A new paragraph at the bottom of a page must have at least two full lines of text. If not, it should begin on the next page.
- c. All paragraphs should be “justified”.

3.1.5 Margins

Margin specifications are meant to facilitate binding and trimming. The stipulated margins for the general text are as follows:

- i. Top edge: 4.0 cm
- ii. Right side: 2.5 cm
- iii. Left side: 4.0 cm
- iv. Bottom edge: 2.5 cm

All information including text headings, footnotes and illustrations should be within these margins.

3.1.6 Spacing

Use 1.5 spacing between lines and paragraphs as well as:

Between Headings and Text.

Text and Headings

Text and Equations

Text and Tables

Appendices, such as Letters, extra data, drawings

3.1.7 Referencing Citation and Caption

- a. For citation, candidates must use the American Psychological Association (APA) style 6th edition. The style should be used consistently throughout the project work. Candidates are advised to refer to the latest conventions of writing from websites.
- b. Equations, figures and tables should have numbers, captions and should be referenced from the text. Unless you have obtained prior permission from your supervisor to submit a separate folio, all maps, photographs, graphs, etc should be embodied in the text. Maps, charts, graphs, diagrams should all be referred to as figures and numbered based on a specific chapter. Sources of information used on compiling equations, figures and tables should be acknowledged. All illustrations should be numbered, titled, and as far as possible, be self-explanatory. Where necessary, each new section should be restarted with a new numbering scheme.

Note: Students can use any software for citation and captions i.e., Mendeley, Endnotes etc.

3.1.8 Headings and subheadings

Headings and subheadings should be numbered as shown in Table 3.1.

Table 3.1 Heading and Subheading

Headings	Font Size	Levels
Cantered, Bold, Uppercase Heading e.g. CHAPTER ONE: INTRODUCTION	14	Level 1
Numbered, Bold, Aligned and Left Capitalized each word except conjunction, proposition (linking words) eg. 1.1 Background Information 1.2 Trends in the Construction Industry	12	Level 2
Numbered, Bold, Aligned Left, Capitalize Each Word Heading e.g. 2.1 Construction Industry	12	Level 2
Indented, bold, aligned left, sentence case heading e.g. 2.1.1 Tanzanian construction industry 2.1.2 International construction industry	12	Level 3
Indented, aligned left, sentence case heading e.g. Alternate between roman numbers and alphabets	12	Level 4

3.1.9 Equations and formulae numbering

The relevant equations and formulae are grouped and numbered sequentially within the respective chapter. For example, equations and formulae found in Chapter three should be numbered Equation 3.1, Equation 3.2, Equation 3.3, and so on. For example, the first equation should be numbered as:

$$I = I_{Ph} - I_{sat} \times e^{\left(\left(\frac{q(v_o + I \times R_{sh})}{(n \times K \times T_{cell} \times N)}\right)\right)^{-1}} - \frac{V_o + I \cdot R_s}{R_p} \quad (3.1).$$

Where:

I- Output current

I_{ph}- Photo current

I_{sat} -Diode reverse saturation voltage

V_o-Output Voltage

R_s-- Series resistance (Representing voltage loss on the way to external connectors)

R_p- Parallel resistance (Representing leakage currents)

k-Boltzmann 's constant

q- Charge on electron

N_s- Number of cells in series

N- Ideality factor

3.1.10 Cover page

The cover page must have the following information:

- a. Institution name (Capital, Centered, font size 16)
- b. Institution logo (Centered)
- c. Department name (Capital, centered, font size 14)
- d. The main title of the project work must be written in CAPITALS (well indented, center) in 14 bold fonts. A sub-title should be written in capital followed by lower case letters.
- e. The full name (Capitalize each word) and registration number (Capital) of the author should be inserted at the center of the title page (12 font size, bold).
- f. The following words must be inserted below the author's name: "*This Project report submitted in partial Fulfilment of the requirements for the award of Bachelor Degree in (insert name of the degree) of the National Institute of Transport*".
- g. Insert the date (month and year) of completion of the project report at the center bottom of the cover page.

Refer to appendix number 1

3.1.11 Title page

The title page must be arranged as follows:

- a. The main title of the project work must be written in CAPITALS (well indented, center) in 14 bold fonts.
- b. The full name (Capitalize each word) and registration number (Capital) of the author should be inserted at the center of the title page (12 font size, bold).
- c. The following words must be inserted below the author's name: *“This Project is report submitted in partial Fulfilment of the requirements for the award of Bachelor Degree in (insert name of the degree) of the National Institute of Transport”*.
- d. Insert the institute name at the center on top of date
- e. Insert the date (month and year) of completion of the project report at the center bottom of the cover page. Refer to appendix number 2

3.2. Final project report

In a single bound volume, Special project report material should be arranged in the following sequence:

- a. Cover page
- b. Title page
- c. Certification
- d. Author's Declaration
- e. Copyright
- f. Acknowledgements
- g. Dedication, if any
- h. List of abbreviations/acronyms used
- i. Abstract
- j. Table of contents
- k. List of tables, figures, illustrations, etc.
- l. List of appendices
- m. Main body
- n. List of references

o. Appendices

3.2.1 Cover page

Refer to 3.1.10

3.2.2 Title page

Refer to 3.1.11

3.2.3 Certification

The undersigned certify that he/she has read and hereby recommend for acceptance by the National Institute of Transport a project report entitled: “***Design and manufacturing of a solar powered Cassava drying machine***” in fulfillment of the requirements for the award of Bachelor degree in (Insert degree program) of the National Institute of Transport. (Justified, 12 font size, times new roman and 1.5 spacing)

.....

Prof. XYZ

(Supervisor)

Date: -----

3.2.4 Declaration

I, **(Insert full name and bold)**, declare that this project report is my own original work and that it has not been presented and will not be presented to any other Institution for a similar or any other degree award. (Justified, 12 font size, times new roman and 1.5 spacing)

Signature -----

3.2.5 Copyright

This project report is copyright material protected under the Berne Convention, the Copyright Act 1999 and other international and national enactments, in that behalf, on intellectual property.

It may not be reproduced by any means, in full or in part, except for short extracts in fair dealings, for research or private study, critical scholarly review or discourse with an acknowledgement, without the written permission of the Deputy Rector– Academic, Research and Consultancy, on behalf of both the author and the National Institute of Transport. (Justify, 12 font size, times new roman and 1.5 spacing)

NOTE: Declaration and Copyright should be written on separate pages.

3.2.6 Acknowledgement

A brief statement of appreciation in recognition of any special assistance rendered to the author during the period of project work should be included. It should be typed in 1.5 spacing and should not exceed one page in length. (Justified, 12 font size, times new roman and 1.5 spacing).

3.2.7 List of abbreviations/acronym

The list should be arranged in an alphabetical order i.e. A-Z

3.2.8 Abstract

This should be concise but comprehensive. It should be 300 words maximum. Essential points to be covered are: the problem investigated, the procedures and methods employed, the important results achieved, contributions made and the major conclusion reached. (Justified, 12 font size, times new roman and 1.5 spacing)

3.2.9 Table of Contents

The table of contents must provide a title and page number for each chapter/section and sub section of the report and should agree with it exactly. Page numbers should be entered against every item in the table of contents and headings should be differentiated using a numbering system. The table of content must be generated automatically. Refer to appendix 3

3.2.10 List of tables, figures, illustrations, etc.

List of tables, figures and other illustrations should be written on different pages. All tables throughout the entire document should be positioned at the center and must have its captions along with citations at the top center. All figures throughout the entire document should be positioned at the center and must have its captions along with citations at the bottom center (Figure 1.1), (Table 1.2) etc.

3.2.11 List of appendices (Centre font size 12, times new roman and 1.5 spacing)

This list should include headings of all appendices along with their page numbers.

3.2.12 Main body

The main body of the project consists of 5 chapters as follows;

- a. Chapter One: Introduction
- b. Chapter Two: Literature review
- c. Chapter Three: Methodology
- d. Chapter Four: Results and discussion
- e. Chapter Five: Conclusion and Recommendation

a. Methodology

This part Provides a detailed description of how the stated objectives of the project will be achieved. Include a detailed outline of methods used in your analysis and validation. This section of the project is very important and provide a complete picture of how the intended work is accomplished.

A student must show clearly the following information:

- i. Methods and tools used to collect and analyze data

The student should explain in details various alternative solutions (at least three alternatives) as per project problem statement and finally with critical reason (s) the student must conclude which one among the three alternatives is the best alternative to solve the stated problem

Note:

(Best alternative analysis Should be shown by using various methods, i.e., Functional tree, decision matrix analysis, Pareto analysis etc. which will enable the student to get the desired outcome.

- ii. Design calculations and material selection procedures

The design calculation must be detailed as per the project title requirement and All materials used in the prototype must conform with the calculations specified as well as area of application (Consider the standards during design calculation and material selection).

- iii. Engineering Drawing / circuit Design (Detail and Assembly drawing)

This must convey all the information necessary for manufacturing a product.

Note: For Title block and Part list standard sample refer to Appendix 4 &5.

iv. Manufacturing process / Circuit implementation

The detailed manufacturing or implementation procedures on how the prototype will be produced must be clearly stated

v. Testing procedures

The student (s) should clearly elaborate in details on how the prototype will be tested so as to answer the main objective and to solve the problem as previously stated

Not only that but also the student (s) should clearly present the result which gained from the design of the prototype and how many experimental replications were carried out on the prototype testing.

Note:

Before prototype testing the parameter (s) of testing must be clearly highlighted

For example

If the aim of the project is plantain pulp slicing- here the designer (student(s)) should specify the parameter to be tested. i.e., **Number of slices per 60s of the machines running time**

Table 3.2 UMBER OF SLICES PER 60s OF MACHINE RUNNING TIME

Number of replications	Number of slices (ripe)	Number of slices (un-ripe)
1.	400	420
2.	380	400
3.	400	375
4.	392	405
Average	393	400

Note that:

- The project must be manufactured in the institute premises unless tools and equipment required to accomplish the project are not accessible in the Institutes workshop, with this mind the student will be required to consult all of the following persons in order to accord the permission: workshop/laboratory manager, project supervisor, project coordinator and head of department (TET).
- The prototype produced must match with the detailed design.
- All specific objectives must be answered in the material and methods section.

b. Results and Discussion

This section of the project that present the findings and outcomes of the project and reason for those particular results in form of text, figure, graphics, tables and charts.

Note: the student (s) should state clearly the reference outcomes based on requirement of the design and compare it with the actual outcomes after testing so as to check for conformity.

i.e. as per above example the requirement of the design was to produce $5\pm0.5\text{mm}$ plantain slices which is the distance between the blade and cutting disc.

TABLE 3.3 CONFORMITY OF SLICE WITH THE STANDARD THICKNESS $5\pm0.5\text{MM}$

Number of replications	RIPE No of slices that conform standard	RIPE No of slices that not-conform standard	UN-RIPE No of slices that conform standard	UN-RIPE No of slices that not-conform standard
1	380	20	390	30
2	368	12	392	81
3	360	40	360	15
4	372	20	380	25
Average	370	23	380.5	19.5

i. Efficiency of the Prototype

Student should state the efficiency of the machine

c. Conclusion, Recommendations

This part of the report should describe the conclusions drawn from the work. This section is particularly important in a final report where it should give a clear description of what is important in the work done, indicating clearly the personal contribution. There should also be a description on how well the initial objectives of the project have been met.

This is to summarise the recommendations and/or improvements for future work arising from the project. It should also discuss the limitations faced in carrying out the project and the scope for further project based on the results.

3.2.13 Reference List

This section begins on a fresh page bearing the heading REFERENCES in capital letters, centered without punctuation. The list of references begins four spaces below the heading and is 1.5 spaced in the same citation but 1.5 spaced between citations. This list must include all the references that candidates have cited in the text of the project. It is recommended that candidates use format as presented in the references format, refer to “*Referencing Citation and Caption*” section

3.2.14 Appendices (Centre font size 12, times new roman and 1.5 spacing)

This should include Gantt chart, Cost estimate, engineering(technical) drawings and all other supplementary information. Appendices should contain detailed data or information that explains the summaries provided in the main text. All appendices must be referred to in the text.

3.3 Proof-reading and Editing

- i. Candidates should ensure that the written project report demonstrates a high standard of proof-reading and copy editing (including attention to layout, spelling, grammar and sentence structure). The report should be checked for accuracy, including references, cross-references and sequences of numbers. Figures, diagrams and tables must be numbered sequentially in the manuscript.
- ii. Candidates should note that, as a general rule, supervisors are not expected to edit a project report. If a project report requires editing, the supervisor(s) may advise the candidate of the need and, if appropriate, give advice on where to obtain help with typographical proof-reading and editing. Candidates are warned that NIT specifically TET is not responsible for the costs incurred for editorial works.
- iii. Candidates are responsible for ensuring that no changes are introduced to the intellectual content in the process of typographical editing.

3.4 Submission of Project Report

- i. Candidates should not submit their project report until their supervisors have had an opportunity to comment on it, and have certified in writing that the project report is ready for submission. A student must now submit three copies of hardbound project report to the project coordinator’s office for presentation.
- i. Student must submit the signed consultation Form (at least 10 forms per semester) along with their project report.

- ii. The title of the work shall be printed in light blue back cover. Candidates may arrange the binding of their project report with any binding firm provided they follow the specifications indicated in these guidelines.

3.4.1 Submission deadlines

The report should be submitted as per coordinator's submission MEMO. To request an extension, a recommendation supported by your supervisor must be made in writing before the original deadline, and must be for a specific length of extension (not more than three days after the deadline). The recommendation must state the grounds for the extension and include reference to appropriate evidence that the supervisor has seen. The grounds must be based on one or more particular circumstances which are clearly beyond the control and which do not constitute a contingency for which provision should have been made.

Note: No project report will be accepted after the deadline.

Examples of such circumstances are:

- i. Medical certificate covering indisposition for a period at which key work was due to be carried out.
- ii. An unexpected bereavement,
- iii. Absence of a supervisor and replacement supervisor at a point when key supervision had been specifically planned despite notice being given to the project coordinator and the coordinator agreeing to provide a replacement.

Examples of circumstances which would not normally be sufficient for an extension:

- i. Loss of a computer disk (backups should be kept);
- ii. Failure of a printer (printing should be done well in advance of the deadline);
- iii. Problems in binding including lack of power (binding should be done well in advance of the deadline)

When deciding whether the grounds are sufficient, the following points will always be considered in relation to the request:

- i. Did the circumstances occur at a key point?
- ii. Has the student kept the supervisor informed of progress so that a valid judgement of the effect of the circumstances can be made?
- iii. Are the circumstances simply “normal problems” for which most students would not expect an extension so that granting an extension in this case would give the student an advantage over other students?

If an extension is granted the project coordinator will ensure that the student is informed in writing with a copy sent to their supervisor and departmental student file, defining the new deadline.

3.5 Project Proposal Presentation

The presentation will be witnessed by Supervisors, students and other departmental staffs in order to have good conduction of project presentations it is recommended that appointed members of teaching staff in the department participate fully in the assessment of the students. Table 3.2 & 3.3 describes the number of presentations, presentation schedule, final report and marks distribution.

Note: To qualify for final presentation, students must attend all presentations as per schedule.

Table 3.4: Proposal Phase

Level 6 & 8	Course Work Accumulation (40%)			Semester Exam (60%)	
	Presentation 1	Presentation 2	Supervisors' Report	Presentation 3	Final Report
Marks Distribution	15%	20%	5%	40%	20%
Remarks	Project Concept presentation (Max. 2 pgs.) Contents; 1. Title 2. Problem statement 3. Objectives - Main objective - Specific Objectives 4. Significance of the project 4. Project cost 5. Gantt chart	Full Project Proposal	How closely the student has been following the project schedule	Project progress up to initial design (engineering drawings of the intended project output and simulations) with regards to comments in presentation 2	Full compiled report.

Table 3.5: Presentation Schedule

Week Number	2 (NTA 08) & 3 (NTA-06)	5	9 (NTA 08) & 11 (NTA-06)	15
Presentation Number	1	Special Project Concept Presentation	2	3
Marks Distribution	15%	15%	20%	40%
Remarks	Project Concept presentation (Max. 2 pgs.) Contents; 6. Title 7. Problem statement 8. Objectives - Main objective - Specific Objectives 4. Significance of the project 9. Project cost 1. Gantt chart	Project Concept presentations (those whose concept note were reject)	Full project proposal presentation	Project progress up to initial design (engineering drawings of the intended project output and simulations) with regards to comments in presentation 2.

3.6 Second Semester Project Execution

There should be three presentations at this stage. The first two will comprise Internal Examiners and the last one which is a final presentation for each project should involve both Internal and External Examiners. There should be not less than four Internal Examiners and one or more External Examiners in each panel. External Examiners should be appointed according to institute regulation. These examiners should be experts in the required area of specialization. Table 3.4 & 3.5 describes the number of Semester 2 Execution Phase, presentation schedule, final report and marks distribution.

Table 3.6: Semester 2 (Execution Phase)

Level 6 & 8	Course Work Accumulation (40%)			Semester Exam (60%)	
	Presentation 1	Presentation 2	Supervisors' Report	Presentation 3	Final Report
Marks Distribution	15%	20%	5%	40%	20%
Remarks	Project Progress Presentation especially the Methodology part (design, drawings, properties of selected materials, circuit designs etc.)	Project Progress Presentation accommodating comments after presentation 1 (Prototype implementation progress)	How closely the student has been following the project schedule	Full project report presentation with regards to comments in presentation 2, Results and Discussion Part (Final project presentation)	Full project compiled report.

Table 3.7: Presentation Schedule

Week Number	2 (NTA 08) & 3 (NTA 06)	9 (NTA 08) & 11 (NTA-06)	15
Presentation Number	1	2	3
Marks Distribution	15%	20%	40%
Remarks	Project Progress Presentation especially the Methodology part (design, drawings, properties of selected materials, circuit design etc.)	Project Progress Presentation accommodating comments after 1 presentation (Prototype implementation progress)	Full project report presentation with regards to comments in presentation 2, Results and Discussion Part (Final project presentation)

CHAPTER FOUR: SUPERVISION OF PROJECTS

4.1 Supervisor

The supervisor plays a critical role in encouraging, guiding and inspiring the project student. As such the supervisor has to clearly understand his/her role to effectively and properly dispense his/her duties and responsibilities in guiding the student. Thus, the supervisor is there not to watch, but to encourage, inspire and guide. The supervisor must have relevant experience, supervisory skills including the “soft skills” (effective communication, relationship management, time management, life skills and conflict resolution) to be effective. In addition, the engineering student must endeavor to produce high quality project results through the special project work. This entails being committed to working hard, taking the supervisor’s advice seriously and adhering to rules and regulations governing student special project research/work.

4.2 Ethics and Code of Practice

- i. Students and supervisors must be aware of and comply with the project ethics and code of practice as used by UNESCO and other Higher Learning Institution in Tanzania.
- ii. Students and supervisors must be aware of ethical issues related to supervision of such projects and Supervisors are reminded to keep a diary of their interaction with students.

4.3 Acts of Plagiarism

- i. Acts of malpractice in the preparation of special project, especially where someone uses another person’s words, ideas, or work and pretends they are his/her own without citing the origin, refers to plagiarism. Such acts constitute cheating which is an academic misconduct as it can result in unearned or undeserved academic award or credit.
- ii. A candidate shall be deemed to have committed an act of Plagiarism if a supervisor, examiner, Head of Department, member of the various committees responsible for checking and certifying compliance to approved publication standards or any other person observes the following:
 - a. One has submitted or presented the work of another person as his or her own;
 - b. There is submission of the same, or substantially the same work more than once at the same or another institution;
 - c. There is fabrication or falsification of results/data;
 - d. There is submission of false records, information or documents;
 - e. There is omission of due acknowledgement of the work of another person;

- f. There is collusion, i.e., when two or more students collaborate to produce the same work submitted by each, without prior formal permission for such collaboration;
- iii. One has used by payment or otherwise a third-party to produce the project work and report in whole or in part.
- iv. All cases of alleged acts of plagiarism shall be referred to the Head of Department who will form a committee and which will have the powers of summoning students and members of staff, as it deems necessary for purposes of examining the case for evidence of plagiarism. Upon completion of the case's examination, the Committee shall submit a report of its findings and recommendations to the Head for approval, prior to any action. Depending on the nature and extent or seriousness of plagiarism committed the following sanctions shall be applicable:
 - a. Rejection of the work (project proposal or project report) or part of the work and therefore the student be required to re-write or re-take the project work;
 - b. Discontinuation from studies;

4.4 Assigning Supervisor(s)

The Head of Department is responsible for recommending supervisor(s) for each candidate, having considered the candidate's project proposal. The Head of Department should carefully consider the proposed supervisor's area of expertise, his/her qualifications and experience before nomination to a supervisory role.

4.5 General Duties and Responsibilities of a Supervisor

- i. The Supervisor is the institutes' agent in ensuring that:
 - a. A student maintains satisfactory progress;
 - b. The student receives adequate advice and encouragement on the project;
 - c. The work being done on the special project is reviewed critically and on a continuous basis.
- ii. From the Institutes point of view, a positive attitude and relationship between the supervisor and the student is essential in order to:
 - a. Ascertain that the student successfully and timely completes the special project work;
 - b. Identify problems, or inadequate work, early enough to avoid unnecessary frustrations to the candidate;
- iii. Since the supervisor is the institutes' agent in ensuring that the student's work attains a satisfactory standard, s/he should be an active member of the Department, who is well acquainted with all the departmental undergraduate special project research regulations.

Where there is more than one supervisor, the principal supervisor must be an active member of the department.

- iv. As far as possible, the supervisor should supervise students only in his/her area of expertise. This would help to ensure high standards of attainment, and to avoid possible embarrassment resulting from rejection of project classified as shallow by examiners.
- v. The supervisor has the responsibility of focusing ahead, and seeing the potential and limitations of the research problem, before the candidate goes too far with the project work. S/he should be capable of foreseeing, through his/her previous experience, whether or not the student will get meaningful data within the specified limited time or whether the proposed project is suitable for undergraduate degree. The supervisor should have this foresight in order to reduce the common problem of students' "wasted effort" to a minimum.
- vi. The supervisor has the responsibility of monitoring the student's progress throughout the special project period. There are various methods through which s/he can do this, including:
 - a. Arranging the timetable which a candidate must follow and submit periodic reports about his/her progress on time. In reviewing reports, supervisors should take the opportunity to advise the candidate on matters of presentation, and if necessary, to make suggestions where modifications are required. If the supervisor does not receive a report from the candidate on his/her project progress, he/she should submit his/her own to the Head of Department.
 - b. Maintaining regular consultations with the candidate (at least once every week) to discuss the project work. Frequent consultations help to make the supervisor aware of new problems in the student's work since the last report and to suggest remedial measures, before it is too late.
 - c. If the supervisor is not satisfied with the performance of at least 50% of project work and 80% of the student's attendance, the project work may be disqualified after the case has been presented to a panel, project coordinator and the head of department.
- vii. Although doing a special project is the responsibility of the student, it is the supervisor's responsibility to ensure that the student submits a report of a standard that is acceptable for the intended degree. In order to enhance the presentation of an acceptable project report, the supervisor should:
 - a. Be accessible, and show interest and enthusiasm in the student's project work;

- b. Read the candidate's draft carefully and critically giving constructive suggestions on how and where the report could be improved;
 - c. Provide timely feedback to the work submitted by the student, as much as possible within one to two weeks of receiving them.
 - d. In the end, read the entire final draft, and satisfy himself/ herself that the special project report is suitable for submission.
- viii. The supervisor's role is to advise and guide the student, and not to write the report for the student such that the work becomes a reflection of the supervisor. The student is fully responsible for the work and its fate.
- ix. If the candidate's sole supervisor is away from the institute (NIT) for more than two consecutive months, a temporary supervisor should be appointed.

4.6 Students' Responsibilities

In terms of responsibilities, the student is expected:

- a. It is the candidate's responsibility to ensure regular consultation with his/her supervisor to discuss the research/ project work and to fill the supervisor student record sheet during each visit.
- b. To follow the timetable assigned by the supervisor and maintain the progress in accordance with the agreed work plan;
- c. To attend regularly the assigned work and place of work and ensure that the department permission is granted for any absence;
- d. To make regular presentations on progress as required by the supervisor / Department.
- e. To make himself / herself aware of, and abide with, the health and safety regulations, for his / her welfare and that of colleagues.
- f. To operate instruments, apparatus, equipment and plants in a way which optimizes safety, precision and longevity;
- g. To exercise the highest level of maturity, honesty and integrity in all curricular and extra-curricular activities.
- h. To abide with the presentation timetable and project submission deadlines.

4.7 Conflict Resolution During Supervision

- i. In the course of the student's research/ project, a situation may develop where one or more of the following may happen:
 - a. A breakdown in communication between the student and the supervisor;
 - b. Personal clashes and conflicts between the student and the supervisor;
 - c. A hostile relationship between the student and the supervisor;
 - d. Refusal by the student to follow the supervisor's advice.
- ii. When such a situation occurs, the supervisor and/or the student should report the problem, in writing to the Head of the Department, with a copy to the Depute Rector Academic Research and Consultancy.

The Head of department should study the nature of the problem and recommend to the Depute Rector Academic Research and Consultancy one of the following actions:

- a. The student be warned, in writing, about his/her weaknesses or conduct (if it is established that s/he is the cause of the problem);
- b. The student be transferred to another supervisor (where possible and necessary);
- c. The supervisor be informed about his/her weaknesses (if it is established that the supervisor is the source of the problem). Depending on the severity of the problem, the supervisor may continue to guide the student, or alternative (b) be followed.
- d. An advisory panel be established by the Head of Department to guide the student, if there is no any other person in the Department who has the expertise to supervise the student; and
- e. Any other reasonable action, including instituting disciplinary proceedings against the student and/or supervisor.

4.8 Supervision Load and Remuneration

- i. The number of students a supervisor can be responsible for depends on his/her rank and other duties in the Department including the teaching load.
- ii. The maximum number of students per supervisor in academic year should not exceed ten (10). This would ensure that candidates are adequately supervised.
- iii. The supervisor should allocate an agreed-upon number of hours per week to each student.
- iv. Every qualified academic staff (with at least a Masters for level 08 and Bachelor degree for Level 06 students) should have the opportunity to supervise final year students.
- v. The Head of Department should monitor the number of students each supervisor has in terms of agreed targets.

- vi. As an incentive and as a token appreciation for the supervisor's work, NIT shall do the following:
 - a. Pay an annual, or else a semester, supervision allowance to supervisors as Council may from time to time prescribe; and
 - b. Give the supervisor a copy of the candidate's project work as his/her permanent record of the work he/she has supervised.
- vii. There should be no link between the payment of the supervision allowance and the candidate's fate (i.e., the supervisor should be paid the allowance even the candidate fails).

4.9 Project assessment mode

All project work should be assessed based on oral defense, written document and presentation mode. Evaluation of the project will be as per curriculum and Institute examination regulations. The assessment form 6a -7d should be used by both external and internal examiners. Failure of any project module (project 1 or project 2) has the same impact like any other modules in a specific semester.

REFERENCES

SUA (2011). *Regulations and Guidelines for Higher Degrees*. 5th Edition [Online] Available at < <http://www.suanet.ac.tz/drpgs/index.html> > [Accessed on 08.01.2016].

SUA (2002). *Guidelines for Preparing Dissertation/Thesis and other Publications*. 2nd Edition [Online] Available at < <http://www.suanet.ac.tz/drpgs/index.html> > [Accessed on 08.01.2016].

UDSM (2013). *General Regulations and Guidelines for Postgraduate Programmes* [Online] Available at < www.udsm.ac.tz > [Accessed on 04.07.2013].

APPENDECES

Appendix 1. Cover page Sample

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING & TECHNOLOGY

DESIGNING AND MANUFACTURING OF A SOLAR POWERED

CASSAVA DRYING MACHINE

Drying Mechanism

Makuru Sangi Mashambo

NIT/BAE/2020/000

**This Project is Submitted in Partial Fulfilment of the Requirement for the award of the
Bachelor Degree in Automobile Engineering of the National Institute of Transport**

NOVEMBER 2020

APPENDIX 2. TITLE PAGE SAMPLE

DESIGNING AND MANUFACTURING OF A SOLAR POWERED CASSAVA DRYING MACHINE

Drying Mechanism

Makuru Sangi Mashambo

NIT/BAE/2020/000

**This Project Report Submitted in Partial Fulfilment of the Requirements for the award of the
Bachelor Degree in Automobile Engineering of the National Institute of Transport**

**National Institute of Transport
NOVEMBER 2020**

APPENDIX 3. TABLE OF CONTENT.

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1.2 The Definition of Agricultural Mechanisation.....	2
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2.1 Introduction	8
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APPENDIX 4. TITLE BLOCK

The following data should be included in the title block

- a) Name of the Institute
- b) Projection Symbol
- c) Scale used
- d) Dimensioning Unit (usually in mm)
- e) Date on which the drawing has been finished
- f) Student Name
- g) Group/Class
- h) Name of person who Checked the drawing
- i) Remarks
- j) Title of the drawing
- k) Drawing number
- l) Sheet format

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	PROJECTION	SCALE:	DRAWN:		REMARKS		
		DIMENSION: mm	GROUP:				
		DATE:	CHECKED:				
	NATIONAL INSTITUTE OF TRANSPORT	TITLE:			DRAWING No.	A4/A3	
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APPENDIX 5. PART LIST

The part-list must have the same width as the title block. The height depends on the number of parts available in assembly drawing.

The following information can be mentioned in the parts lists:

- a) Part reference number
- b) Part name
- c) Material
- d) Dimensions
- e) Drawing

number

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	ITEM	NAME	PART No.	MATERIAL	STANDARD OR DIMENSION	DRAWING No.
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		DIMENSION: mm		GROUP:		
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NATIONAL INSTITUTE OF TRANSPORT			TITLE:			DRAWING No. A4/A3

APPENDIX 6 A.

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROGRAM NAME

Academic Year:

Semester: 1

Registration Number. 1.....

2.....

3.....

NTA.....

Project Title:

PROJECT- CONCEPT NOTE- PRESENTATION
ASSESSMENT (100%)

S/N	Area of Assessment	Max. Score	Act. Score	Weighted Scores (15%)
1.	Quality of concept note as per project guideline			
	i. Title of project	10%		
	ii. Problem statement	25%		
	iii. Setting Objectives	25%		
	iv. Significance of the project	10%		
	v. Cost estimates	15%		
	vi. Gantt Chart	5%		
2.	Presentation Skills			
	i. Appearance	2%		
	ii. Confidence	2%		
	iii. Interaction with the audience	2%		
	iv. Material Flow	2%		
	v. Time adherence	2%		
TOTAL		100%		

Remarks:

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Examiner's name: Sign.....Date.....

APPENDIX 6 B.

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROGRAM NAME

Academic Year:

Semester: 1

Registration Number. 1.....

2.....

3.....

NTA.....

Project title:

PROJECT I MINI PRESENTATION ASSESSMENT (100%)

S/N	Area of Assessment	Max. Score	Act. Score			Weighted Score (20%)	
1.	Quality of proposal as per project guideline						
	i.	Project title	5%				
	ii.	Problem statement	15%				
	iii.	Setting Objectives	10%				
	iv.	Significance of the project	5%				
	v.	Literature review	20%				
	vi.	Methodology	25%				
	vii.	Cost estimates	5%				
	viii.	Project Schedule	5%				
2.	Presentation Skills						
			1	2	3		
	i.	Appearance	2%				
	ii.	Confidence	2%				
	iii.	Interaction with the audience	2%				
	iv.	Material Flow	2%				
	v.	Time adherence	2%				
TOTAL		100%					

Remarks:

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Examiner's name: Sign..... Date

APPENDIX 6 C.

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROJECT SUPERVISORS' ASSESSMENT FORM

PROGRAM NAME

Academic Year:

Semester: 1

Registration Number. 1.....

2.....

3.....

NTA.....

Project title:

S/No	Area of Assessment	Max. Score	Actual Score	Weighted Score (5%)
1.	How closely the student followed the events of the project calendar	5%		
2.	Ability of the candidate to work independently with little guidance	5%		
3.	Problem statement and its significance (is the project worth?)	20%		
4.	Project objectives/s	10%		
5.	Literature review	20%		
6.	Methodology	30%		
7.	Quality of the project proposal report	10%		
TOTAL		100%		

Remarks:

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Supervisors' Name

Signature

APPENDIX 6 D. ASSESSMENT FORM

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROGRAM NAME

Academic Year:

Semester: 1

Registration Number. 1.....

2.....

3.....

NTA.....

Project title:

PROJECT I - END OF SEMESTER PRESENTATION ASSESSMENT (100%)

S/N	Area of Assessment	Max. Score	Act. Score			Weighted Score (40%)
1.	Quality of proposal as per project guideline					
	i. Project title	5%				
	ii. Problem statement	5%				
	iii. Setting Objectives& Significance of the project	5%				
	iv. Literature review	15%				
	v. Initial design calculations and material selection	30%				
	vi. Drawings (conceptual drawing, Circuit, sketches, Block diagram, Simulation)	25%				
	vii. Cost estimates and Project Schedule	5%				
2.	Presentation Skills					
		1	2	3		
	i. Appearance	2%				
	ii. Confidence	2%				
	iii. Interaction with the audience	2%				
	iv. Material Flow	2%				
	v. Time adherence	2%				
TOTAL		100%				

Remarks:

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Examiner's name: Sign..... Date

APPENDIX 7 A. ASSESSMENT FORM**NATIONAL INSTITUTE OF TRANSPORT****DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY****PROGRAM NAME****Academic Year:****Semester:** 2**Registration Number. 1**.....**2**.....**3**.....

NTA.....

Project title:**PROJECT II - PROGRESS PRESENTATION ASSESSMENT (100%)**

S/N	Area of Assessment	Max. Score	Act. Score			Weighte Score (15%)
1.	Quality of project document as per project guideline					
	i. General Introduction	5%				
	ii. Objectives	5%				
	iii. Design calculation and material selection	25%				
	iv. Drawings (tolerance, surface finishing, circuits, development etc.) and simulations.	30%				
	v. Production Procedures	15%				
	vi. Linkage between Project Objectives and Methodology	10%				
2.	Presentation Skills		1	2	3	
	i. Appearance	2%				
	ii. Confidence	2%				
	iii. Interaction with the audience	2%				
	iv. Material Flow	2%				
	v. Time adherence	2%				
TOTAL		100%				

Remarks:

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Examiner's name: Sign..... Date

APPENDIX 7 B. ASSESSMENT FORM.

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROGRAM NAME

Academic Year:

Semester: 2

Registration Number. 1.....

2.....

3.....

NTA.....

Project title:

PROJECT II - MINI PRESENTATION ASSESSMENT (100%)

S/N	Area of Assessment	Max. Score	Act. Score	Weighted Score (20%)			
1.	Quality of project document as per project guideline						
	i. General Introduction	5%					
	ii. Design calculations and material selection	15%					
	iii. Drawings (tolerance, surface finishing, development, PCB etc)	30%					
	iv. Progress made towards implementing the prototype	40%					
2.	Presentation Skills				1	2	3
	i. Appearance	2%					
	ii. Confidence	2%					
	iii. Interaction with the audience	2%					
	iv. Material Flow	2%					
	v. Time adherence	2%					
TOTAL		100 %					

Remarks:

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Examiner's name: Sign..... Date

APPENDIX 7 C. ASSESSMENT FORM

NATIONAL INSTITUTE OF TRANSPORT



DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

PROJECT WORK II SUPERVISORS' ASSESSMENT FORM

PROGRAM NAME

Academic Year:

Semester: 2

Registration Number. 1.....

2.....

3.....

NTA.....

Project title:

S/No	Area of Assessment	Max. Score	Actual Score	Weighted Score (5%)
1.	How closely the student followed the events of the project calendar	5%		
2.	Ability of the candidate to work independently with little guidance	5%		
3.	Project objectives/s achievements	20%		
4.	Methodology	20%		
5.	Results and discussion	20%		
6.	Overall quality of the project-output	20%		
7.	Quality of the project report	10%		
TOTAL		100%		

Remarks:

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Supervisors' Name

Signature

APENDEX 7 D. ASSESSMENT FORM**NATIONAL INSTITUTE OF TRANSPORT****DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY****PROGRAM NAME****Academic Year:****Semester: 2****Registration Number. 1**.....**2**.....**3**.....**NTA.....****Project title:****Project II- Semester Project Execution Assessment (100%)**

S/N	Area of Assessment	Max. Score	Act. Score			Weighted Score (40%)
1.	Quality of project document as per project guideline					
	i. General Introduction	5%				
	ii. Design calculation and Material Selection	10%				
	iii. Drawings (tolerance, surface finishing, development etc)	20%				
	iv. Prototype Implementation	30%				
	v. Prototype testing Procedures	10%				
	vi. Results, Discussion and Conclusion	15%				
2.	Presentation Skills		1	2	3	
	i. Appearance	2%				
	ii. Confidence	2%				
	iii. Interaction with the audience	2%				
	iv. Material Flow	2%				
	v. Time adherence	2%				
TOTAL		100%				

Remarks:

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Examiner's name: Sign..... Date

APPENDIX 8.

Supervisor / Student Consultation Form

This form has to be submitted by a student each time he/she meets the supervisor. The student has to consult his/her supervisor at least 10 times in a semester.

CANDIDATE – SUPERVISOR MEETING RECORD SHEET

DEPARTMENT OF TRANSPORT ENGINEERING AND TECHNOLOGY

Date:

--

Candidate Reg. No:

--

Supervisor's Name:

Discussed:

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Suggested date of Next meeting:

Candidate Signature (representative):

Supervisor Signature: